Authentication Module – Functional Workflow

Project: TaskFlow – Intelligent Task Allocation & Tracking System
Module: Authentication · **Child Modules:** Login, Account Recovery, MFA Security Setup, Sessions and Tokens
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1. Module Overview

The Authentication module is the front door of TaskFlow. It lets registered users sign in with their email (or username) and password, adds a second step with an authenticator app (two-step verification), lets people who forgot their password reset it themselves with that app, and keeps their session alive safely until they sign out.

2. Purpose / Business Value

- Only active, registered users can enter TaskFlow; inactive or deleted accounts are refused (BRD Screen 1, requirement 5).
- A stolen password alone is not enough: every user must set up an authenticator app before they can use the application (two-step verification is required by default).
- Users who forget their password can recover it from the login page without waiting for an administrator, using the code from their authenticator app. No password or code is ever sent by SMS or email.
- Sessions end automatically when a password is changed or reset, or when an administrator resets someone's two-step verification, so an old session cannot outlive a security change.
- After sign-in each user lands on the dashboard that matches their role and permissions (BRD Screen 1, requirement 3).

3. Who Can Access This Module

Role	Can See Module	Can Use Module	Access Level (Own / Team / Organization)
Administrator	✅	✅	Own (sign-in, recovery, own security setup); Organization for "Reset two-step verification" of another employee
Manager	✅	✅	Own
Employee	✅	✅	Own
Custom Role	✅	✅	Own; "Reset two-step verification" only if granted (Configurable)

The sign-in, account recovery and security setup screens are open to everyone (they need no permission key). The security setup screen also needs the user to be signed in and to still have two-

step verification to set up.

4. Screens in This Module

#	Screen Name	Menu Path	Purpose	Visible To
1	Login (Sign in)	Opening page `/login` (no menu)	Enter email and password; then the 6-digit authenticator code when two-step verification is on	Everyone (not signed in)
2	Two-step verification (second step of Login)	Shown on `/login` after a correct password	Enter the code from the authenticator app	Users whose two-step verification is on
3	Forgot your password? (Account Recovery)	Login → **Forgot password?** (`/account-recovery`)	Verify with an authenticator code, then choose a new password	Everyone (not signed in)
4	Let's secure your account (Security Setup)	Opens automatically after sign-in (`/security-setup`)	Confirm your details, connect an authenticator app, save backup codes	Signed-in users who have not set up two-step verification yet
5	Sign out button	Header → account area → **Sign out** icon	End the current session	All signed-in users

5. Child Modules

#	Child Module	Purpose	Visible To	Document
01.1	Login	Password sign-in, two-step code at sign-in, redirect to the right dashboard	Everyone (not signed in)	`01.1-Login.md`
01.2	Account Recovery	Reset a forgotten password with the authenticator app, without an administrator	Everyone (not signed in); works only for accounts with two-step verification on	`01.2-Account-Recovery.md`
01.3	MFA Security Setup	Mandatory first-time setup of the authenticator app and backup codes; administrator reset of an employee's two-step verification	Signed-in users with setup pending; reset: users with `employees.mfa.reset`	`01.3-MFA-Security-Setup.md`
01.4	Sessions and Tokens	Keep the session alive, check it regularly, end it on sign-out or after security changes	All signed-in users	`01.4-Sessions-and-Tokens.md`

The child documents are in the folder `01-Auth/` next to this file.

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6. Permission Summary (Module Level)

Permission Key	Description	Administrator	Manager	Employee	Custom Role
(none – public)	Sign in, two-step code at sign-in, account recovery, session renewal, sign-out	Always On	Always On	Always On	Always On
`profile.view`	View my profile (used by Security Setup step 1 to load details)	Always On	Always On	Always On	Always On
`profile.edit`	Edit my personal information (Security Setup step 1)	Always On	Always On	Always On	Always On
`profile.mfa.manage`	Manage my authenticator: set up two-step verification and regenerate backup codes	Always On	Always On	Always On	Always On
`employees.mfa.reset`	Reset two-step verification of an employee (Employee Management)	Granted	Denied	Denied	Configurable

7. Related Modules

- **Dashboard** – after sign-in the user is sent to the Admin, Manager or personal dashboard (decided by role and the `dashboard.view.*` permissions).
- **Profile** – the user's own details and MFA endpoints live under `/users/me`; changing the password there needs an authenticator code and ends all sessions.
- **Management (Employee Management)** – administrators create accounts, activate/deactivate them (inactive accounts cannot sign in) and reset an employee's two-step verification.
- **Settings (Role Configuration)** – role and permission changes reach signed-in users on their next request through the session check.
- **Communication (Notifications)** – system notifications are sent after a password recovery and after an administrator MFA reset.
- **Legal** – the legal footer is shown on the Login and Account Recovery screens.

8. Expected Outcome

Every registered, active user can sign in with their password and authenticator code and lands on the dashboard for their role. New users are guided through securing their account before they can do anything else. A user who forgets their password can reset it alone in a few minutes, and any password change, reset or administrator MFA reset immediately ends old sessions, so the organization's data stays protected.

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